



OCULUS
PENTACAM® /
PENTACAM HR®



INSTRUCTION MANUAL

System for measuring and analysing the front part of the eye

Notes on this instruction manual

Thank you for your purchase and the trust you have placed in this OCULUS product. The Pentacam® / Pentacam HR® has been manufactured and tested according to strict quality criteria. You have selected a modern and well-engineered product.

To ensure safe operation, it is essential that you use the device correctly. For this reason you should familiarise yourself thoroughly with the contents of this instruction manual before operating the device. In particular, pay attention to the safety instructions.

- The operating concept of both the Oculus Pentacam and Pentacam HR is basically identical.
- This instruction manual describes how to manage patient data, the default settings of the Pentacam program and the measuring process.
- Additional functions pertaining to the Pentacam HR (high resolution) are indicated accordingly.
- The Pentacam reference manual contains information supplementing the description of the operating concept.

Due to ongoing development, the diagrams shown may depict minor changes to the actual device delivered.

If you have any queries or would like additional information about your device, do not hesitate to call or send us a fax. Our service team will gladly assist.

OCULUS Optikgeräte GmbH



OCULUS is certified according to DIN EN ISO 9001:2000 and 13485:2003, setting high standards of quality where development, manufacture, quality assurance and service regarding the entire range of products are concerned.

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1 Scope of Delivery

Component	Order number
Pentacam	70700
Pentacam HR	70900
■ Pentacam mounted on an x-y-z mobile unit	70518
■ Head and chin rest	
■ Plate for mounting on a refraction unit or table (Dimensions 28 cm x 36 cm)	70517-01-006
■ Power supply, input 100 V-240 V, output 24 V, 2.1 A, Hitron HES 49	05150150
■ USB cable	70900-07-001
■ Licence key (dongle)	70740
■ Testing log, electrical safety	
■ Instruction Manual	G / 70700 (HR): G / 70900
■ Dust protection cover	60100-5/1
■ Paper for chin support	65313
■ Software package and books	
■ Windows(tm) software package and operating instructions	
■ Dark sheet	027070000006

Standard software module	Order number
■ Software module: Pachymetry	70721
■ Software module: Densitometry	70722
■ Software module: 3D chamber analysis	70723
■ Software module: Corneal topography, front and rear surface	70724

Optionally available modules	Order number
■ Colour printer	70520
■ Simulation program for adjusting contact lenses	70726
■ Simulation program for fitting phakic IOLs, incl. age simulation (Pentacam HR only)	70928

[We reserve the right to change the scope of delivery in line with ongoing technical development]

2 Safety instructions

2.1 Safety instructions concerning organisation

The law requires that the manufacturer expressly informs the user about safety aspects concerning the handling of the Pentacam® / Pentacam HR®. This chapter contains a summary of the most important points regarding these safety matters.



Attention

Do not use the device if you have not understood the instruction manual.

- Make sure to keep this instruction manual in a safe place and available to operating personnel at all times.
- Observe the legal regulations with regard to accident prevention.



The Pentacam is a high-quality technical product. To ensure that it functions correctly and safely we recommend the following: Have the Pentacam checked by our service personnel or an authorised dealer every two years. If an error occurs which you are unable to correct, label the device as "out of order" and contact our service department.

2.2 Pictogram definitions



Attention

Identifies a potentially dangerous situation which may cause minor injury or damage to property.



Note

Instructions for use, and useful or important information.



Note

Identifies important information about the product or on how to use it, which requires special attention.

2.3 Safety instructions regarding the use of the Pentacam

Note

Before you use the Pentacam for the first time you must be instructed either by us or an authorised dealer.



Attention

No modifications may be made to this device without the permission of the manufacturer.



- Only operate the Pentacam using original accessory parts supplied by us, and when the device is in technically correct working order.
- Never operate a Pentacam that is damaged. Should such a situation arise, contact the supplier.
- Do not operate the devices included in the delivery in areas where explosions may occur, where there are inflammable anaesthetics, volatile substances such as alcohol or petrol nearby.
- Do not operate or store the Pentacam in damp rooms.
- Keep the Pentacam away from water that may drip, surge or splash and make sure that no liquids can enter the Pentacam. Do not place any containers with liquid either close to or on the Pentacam.
- When cleaning, use a damp cloth and make sure that no liquid enters the Pentacam.
- Only operate the Pentacam in rooms used for medical purposes if VDE 0100 installation procedures have been observed.
- Do not force any plug connections.
If you are unable to make a plug connection, check whether the plug fits the socket.
If you detect damage to the connection, you should let our service personnel repair the damage.
- When disconnecting electrical connections, pull on the respective plug instead of the cable itself.
- Do not cover the ventilation openings.

Attention

Any additional equipment that is to be connected to either analogue or digital interfaces must verifiably fulfil appropriate EN or IC specifications. Furthermore, all configurations must comply with the IEC 601 - 1 system norm.





Attention

If you connect the Pentacam with non-medical equipment (e.g. data processing equipment) to a medical electrical system, doing so must not cause patient-relevant safety levels to fall below IEC 601 - 1. If making this connection leads to the leakage current threshold being exceeded, protective measures including a circuit breaker must be in place.

3 Proper Usage

The OCULUS Pentacam is a measuring device for examining the front part of the eye and may only be used for the purpose described in this instruction manual.

The OCULUS Pentacam is designed for use in clinics and by eye specialists and opticians. It must be used in the area designated for carrying out examinations.

Only operate the Pentacam using original accessory parts supplied by us, and when the device is in technically sound condition.

You must use the power supply belonging to the device in order to operate it. There is no other method of connecting the device possible.

OCULUS guarantees that the Pentacam will function correctly only if no extra software or programs (except for Microsoft updates and, if required, patient data management) are installed on the laptop or desktop PC.

For this reason the device may only be operated by personnel instructed to do so, who, with appropriate training, knowledge and practical experience, are able to ensure proper handling of the device.

4 Transportation and Disposal

4.1 Transport and storage instructions



Note

Great care must be taken when transporting the Pentacam from one location to another.

- ➔ Avoid operating or storing the device next to heaters/radiators or in damp areas.
- ➔ Check the Pentacam for damage every time it has been transported.

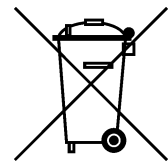
The requirements for transporting and storing the device according to IEC 601 - 1 are as follows:

Ambient temperature:	-40 °C to +70 °C
Relative humidity, including condensation:	10% to 100%
Air pressure:	500hPa to 1060hPa

These levels apply in conjunction with the use of appropriate packaging and for a maximum period of 15 weeks.

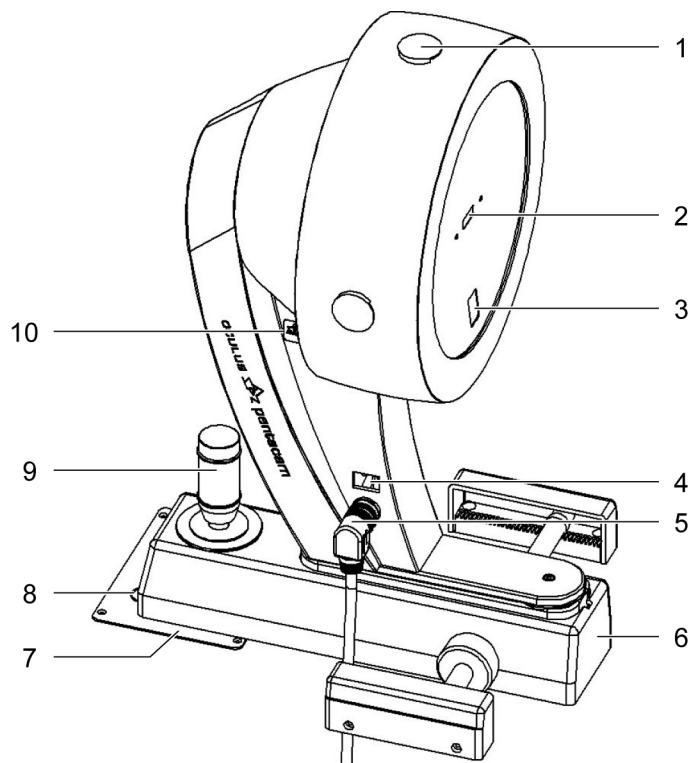
4.2 Disposal of used devices

In accordance with Directive 2002/96/EC of the European Parliament and the Council of 27 January 2003, and in accordance with German law governing the circulation, return and environmentally friendly disposal of used electrical and electronic devices, such appliances must be recycled and may not be discarded as household waste.



5 Device Description

5.1 Overview of device components



1 Ventilation opening

2 Measuring window

3 Camera opening

4 On/off switch with indicator light

5 Y-cable connector

6 Cross slide

7 Sliding plate

8 Circular markings on sliding plate

9 Joystick

10 Nameplate

Fig. 5-1: Device components

5.2 How the Pentacam works

While rotating around the eye, the Pentacam creates Scheimpflug images of the front part of the eye through varying axes. The images which this rotation produces are the basis for calculating height data which is used to extrapolate all other results and to create a 3D model.

The entire measuring process takes less than two seconds. Up to 25,000 (HR: 138,000) genuine height values are measured and analysed. At the same time any eye movement is recorded, which the 3D model takes into account.

When the PC has received the appropriate data record it calculates a 3D model of the front part of the eye. The 3D model provides the basis for all subsequent analysis.

The topography of the front and rear surfaces of the cornea and the pachymetry are calculated and displayed from limbus to limbus for the entire surface of the cornea.

An analysis of the anterior chamber provides the basis for calculating the chamber angle, chamber volume and chamber depth.

A rotatable 3D model displays the front and rear surfaces of the cornea, the iris and the lens. Densitometry of the lens automatically produces quantified values.

The Scheimpflug images created during an examination are digitalised in the measuring head and transmitted to the PC.

Colour images displayed on the screen show the results of the measurement.

A quality specification (QS) lets you gauge the quality of the measurement taken.

Attention

OCULUS Optikgeräte GmbH shall not be liable in any form for further use of the data recorded by a Pentacam and for the evaluations it has calculated.



6 Operation

6.1 First-time operation

Attention

Before you use the device for the first time, the OCULUS Pentacam examination area must be installed and connected by our service personnel or an authorised specialist.



- ➔ Wait approx. 3-4 hours after transport before operating the Pentacam. If the Pentacam was stored in a cold room or vehicle during the cold time of the year, a significant change in temperature may cause condensation to appear on optical parts in the Pentacam.

6.2 Daily operation

If you move the Pentacam to another location, you must position it so that direct light cannot influence measurements. As a rule you must ensure that any examination is conducted without interference from reflections. For this reason you should operate the Pentacam in a dark room.

As it is an optical device, the Pentacam demands careful handling. Avoid jolting, blows, contamination or high temperatures.

- ➔ The first step is to switch on the PC or laptop.
- ➔ Then switch on the Pentacam at the power supply (position ON). The LED on the switch lights up in green.

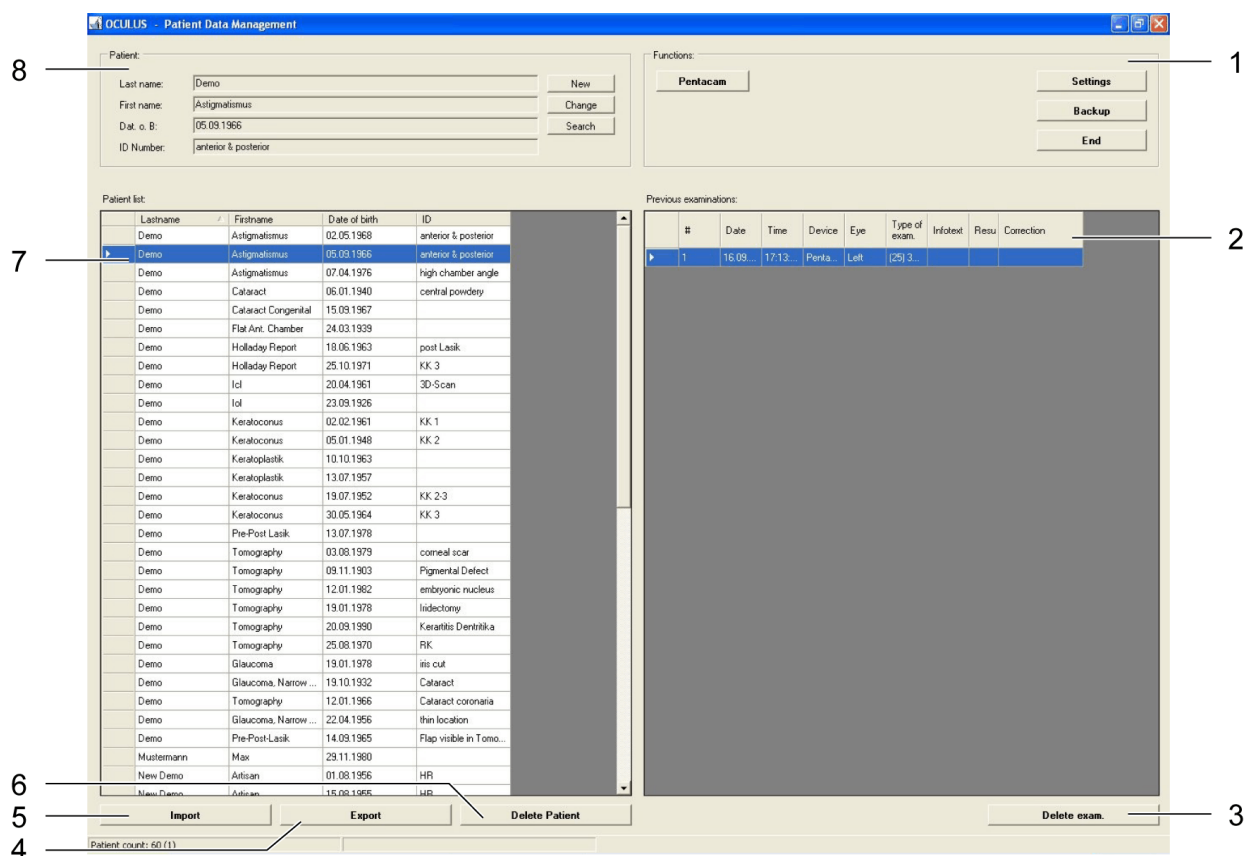
6.3 Switching the Pentacam off

- ➔ Close the Pentacam program and Patient Data Management.
- ➔ Shut down the Windows operating system.
- ➔ Switch off the Pentacam at the power supply.

7 Patient Data Management

7.1 Starting Patient Data Management

After you have switched on the PC, it first loads the operating system. It then displays the user interface for "Patient Data Management".



1 "Functions" group box

2 Previous examinations

3 [Delete exam.] button

4 [Export] button

Fig. 7-1: Patient Data Management

5 [Import] button

6 [Delete Patient] button

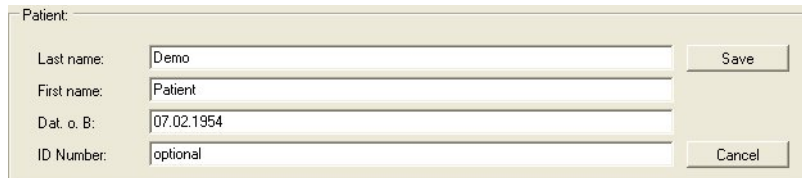
7 Patient list

8 "Patient" group box

- ➔ To enter the Pentacam program, you must either enter a new patient (8) into the "Patient" group box or select an existing patient (2) from the list.

7.2 Entering new patients

- To enter a new patient in Patient Data Management, click [New].
- Enter the patient's last name, first name and date of birth completely in the patient window (8).



The image shows a 'Patient' window with the following fields and values:

Field	Value
Last name:	Demo
First name:	Patient
Dat. o. B.:	07.02.1954
ID Number:	optional

Buttons: Save, Cancel

Fig. 7-2: Entering patients

- Optionally you can enter an ID number for the patient.
- To accept the entries you have made, click [Save]. The patient you have just entered now appears in the patient list.
- Select this new patient from the patient list and start the Pentacam program ([sect. 7.4, page 10](#)).

7.3 Selecting existing patients

The patient list on the left-hand side of the screen displays all previously examined patients in alphabetical order.

- Click [Search] to quickly find the patient you require from the list.
- Enter the patient's name or the first letter of the name in the "Last name" field.
- Optionally you can search for the patient by entering his or her ID number, assuming that patient was assigned one when first entered.
- Click the appropriate entry in the list to transfer that patient's name to the patient window. This also brings up a list of any previous examinations for that patient in the examination window (bottom right-hand side).

Extended patient search: [Extended] checkbox

- Activate the [Extended] checkbox.

The screen displays additional search parameters which reference previous examinations, for example. The procedure is similar to entering a new patient name.



The image shows an 'Extended' search window with the following fields and values:

Field	Value
Last name:	
First name:	
Dat. o. B.:	13.08.2007
ID Number:	
Exam. date:	13.08.2007
Device:	
Eye:	
Type of exam.:	
Infotext:	
Result:	

Buttons: Quit Search, Extended (checked), Invert result

Fig. 7-3: Extended search

7.4 Starting the Pentacam program

Switching from Patient Data Management to the Pentacam program:

- ➔ After you have selected a patient: Start the Pentacam program by clicking [Pentacam] (fig. 7-1, page 8).
- ➔ Alternately, you can start the program by double clicking the patient you have selected.

8 Pentacam Program



Note

The Pentacam-Software is not to be used to dictate therapy without professional correlation with other clinical findings and diagnostic tests.

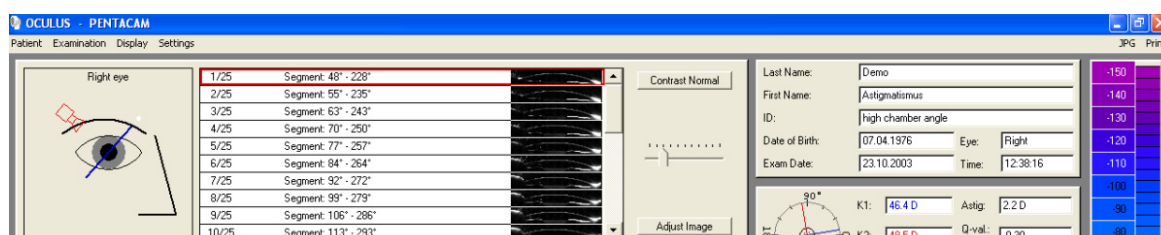


Fig. 8-1: Pentacam program menu (upper section)

Loading previous examinations

- ➔ Select the menu item [Examination] and click [Load].
The dialog box "Load Examination" appears.
- ➔ Make a selection by clicking the required examination.
- ➔ Confirm your selection by clicking [OK], or by double clicking.
The Pentacam program will load the examination you have selected.

Starting a measurement

- ➔ Select the [Examination] tab and click [Scan].
The blue slit light is activated and the Scan menu (sect. , page 10) appears.



Note

As this instruction manual focuses on the Pentacam operating concept, the description of Pentacam program functions is limited to starting the measuring process and loading previous examinations. The Reference manual contains detailed information on the functions of the Pentacam program.

9 Scan Menu

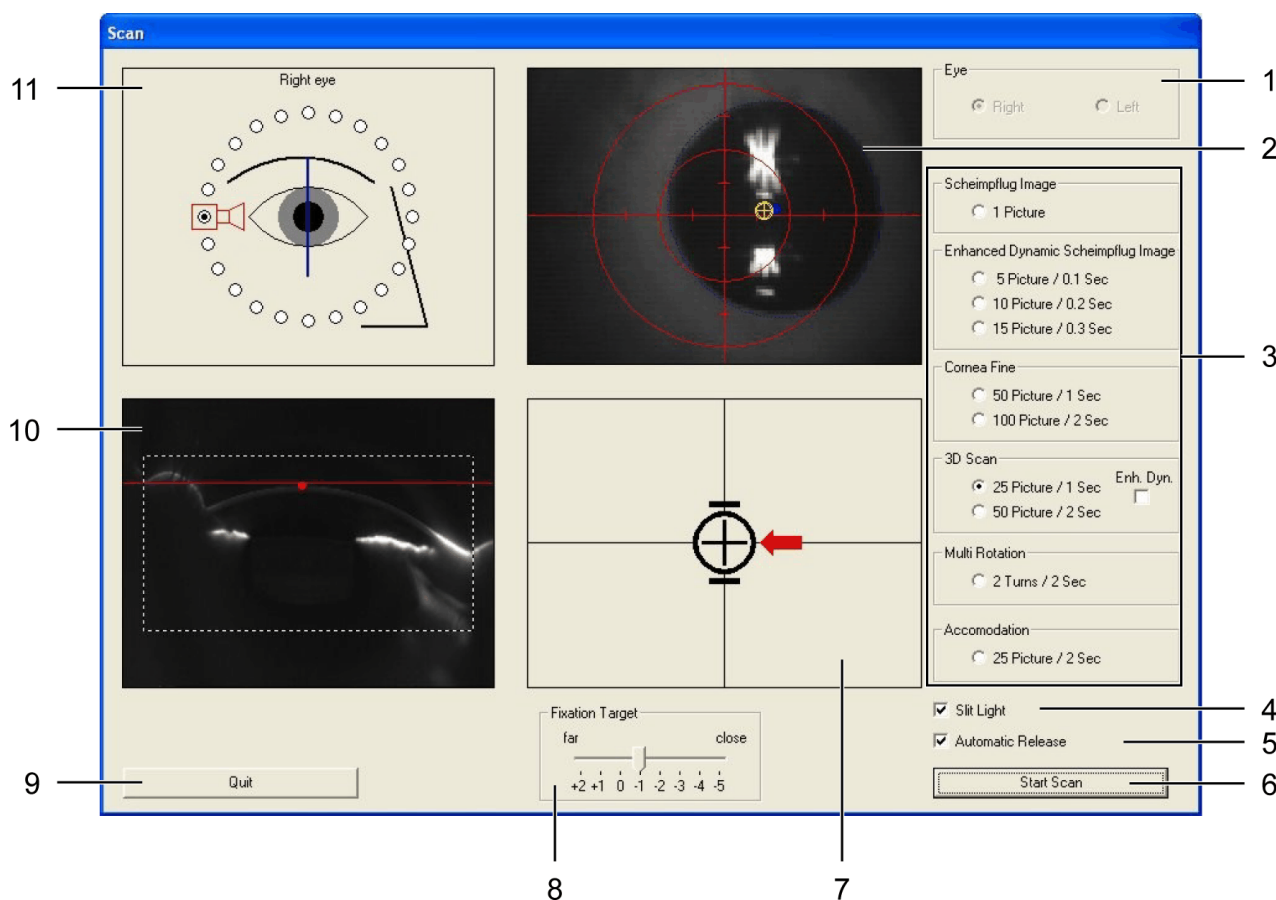
Note

Switching from the Pentacam program to the Scan menu:

- In the Pentacam program (fig. 7-1, page 8) select the menu item [Examination] and click [Scan].



9.1 Screen layout



- | | |
|--------------------------------|-----------------------------|
| 1 "Eye" field | 7 Adjustment window |
| 2 Pupil image | 8 Fixation Target (HR only) |
| 3 "Image Options" area | 9 [Quit] button |
| 4 [Slit Light] checkbox | 10 Scheimpflug Image |
| 5 [Automatic Release] checkbox | 11 "Orientation" field |
| 6 [Start Scan] button | |

Fig. 9-1: "Scan" screen ("HR" model)

- The eye currently being examined is detected automatically and is displayed in the "Eye" field (1).
- The Pupil Image (2) shows the vertical / horizontal positioning of the Pentacam. The object is to position the yellow dot marking the apex of the cornea in the centre of

the crosshairs. The blue dot marks the centre of the pupil, while the blue ring frames the pupil.

- You can set the type of image required for the respective examination in the "Image Options" area (3) ([sect. 9.1.1, page 12](#) and [sect. 9.1.2, page 13](#)).
- Click the [Slit Light] checkbox (4) to activate/deactivate the blue light for illuminating the eye.
- Click the [Automatic Release] checkbox (5) to activate automatic measurement.
- Clicking the [Start Scan] button (6) activates manual measurement. You can also use the Return key.
- Inside the adjustment window (7), there are arrows showing you the direction in which you must move the Pentacam to activate automatic measurement (Automatic Release).
- The "Fixation Target" setting (8) is a parameter used only on the Pentacam HR ([sect. 9.1.2, page 13](#)).
- Click the [Quit] button (9) to abort the measurement.
- The Scheimpflug image (10) shows the distance between Pentacam and patient. The objective of this setting is to move the red dot on the front surface of the cornea so as to coincide with the red line.
- The "Orientation" field (11) shows the respective position of the camera and the eye, which is currently being examined.

9.1.1 Scheimpflug image settings

You can set the number and type of images required for the respective examination in the "Image Options" box (3).

"Scheimpflug Image" group box

- If you activate this option, the camera records only one Scheimpflug image. You can freely select the camera position you require by clicking the white rings in the "Orientation" field (11).

"Enhanced Dynamic Scheimpflug Image" group box

- Use this option to make the camera record either 5, 10 or 15 Scheimpflug images, with the camera remaining in the same position. Image averaging is carried out on the recorded images to minimise background noise. Only one Scheimpflug image is displayed. You can freely select the camera position you require by clicking the white rings in the "Orientation" field. This type of image is suitable for a purely densitometric assessment of the lens.

"3D Scan" group box

- Use this option to select how many images you want the camera to record per scan. The difference is in the duration of the examination and the number of measuring points that are evaluated. A scan comprising 50 images takes longer, but assuming good prior fixation of the patient it is the most accurate. This type of examination is used for evaluating the cornea and anterior chamber.

9.1.2 Parameters applicable only to the Pentacam HR

Note

The parameters described below apply only to the Pentacam HR.



The Pentacam HR has additional options for selecting image types. You can find these exclusive parameters in the "Image Options" box (3).

"Cornea Fine" group box

- Select this option for a more detailed image of the cornea.
The camera does not capture the deeper-lying layers of the front part. You can select 50 Scheimpflug images for a recording time of one second, or 100 Scheimpflug images for a recording time of two seconds.

"Multi Rotation" group box

- If you select this option, the camera will record Scheimpflug images from 50 different positions while performing a full rotation twice around the eye.

"Accommodation" group box

- If you select this option, the camera will take a total of 50 Scheimpflug images. While the camera is recording the images, the "Fixation Target" shifts constantly from -5 dpt to +2 dpt. The camera records the Scheimpflug images from a pre-selected camera position.

[Enh. Dyn.] checkbox in the "3D Scan" group box

- Activating the "Enh. Dyn." function prolongs the exposure time per Scheimpflug image. The advantage is an accurate representation of phakic IOLs. If you select this recording mode, colours and evaluations are neither calculated nor displayed.

"Fixation Target" slider

- Use of the "Fixation Target" enables a better fixation of the patient. For this, the active "Fixation Target" (8), i.e. the LED blinking red in the middle of the blue slit, can be shifted in steps of 0.5 dpt. The object is to offset defects in the patient's vision and ensure a simpler method of fixation.

9.2 Information for recording Scheimpflug images

Type of examination	Examination mode	Images	Automatic measurement	Notes
Topography	3D scan	25-50	Yes	
Pachymetry	3D scan	25-50	Yes	
Analysis of the anterior chamber	3D scan	25-50	Yes	Do not apply mydriatic drops.
Artificial lenses (general)	Enhanced dynamic (for HR in the "3D scan" examination mode)	15	Yes	If the pupil is insufficiently dilated, apply mydriatic drops Use 3D scan for measurements.
Measuring functions	3D scan	25-50	Yes	If the pupil is insufficiently dilated, apply mydriatic drops
Densitometry	3D scan Enhanced dynamic	25-50 5-15	No	Use the same number of images to enable a progress check.

Specific instructions for the Pentacam HR

Type of examination	Examination mode	Images	Automatic measurement	Notes
IOLs, ICLs, PIOLs	3D scan for PIOLs, possibly longer exposure	25-50	Yes	If the pupil is insufficiently dilated, apply mydriatic drops

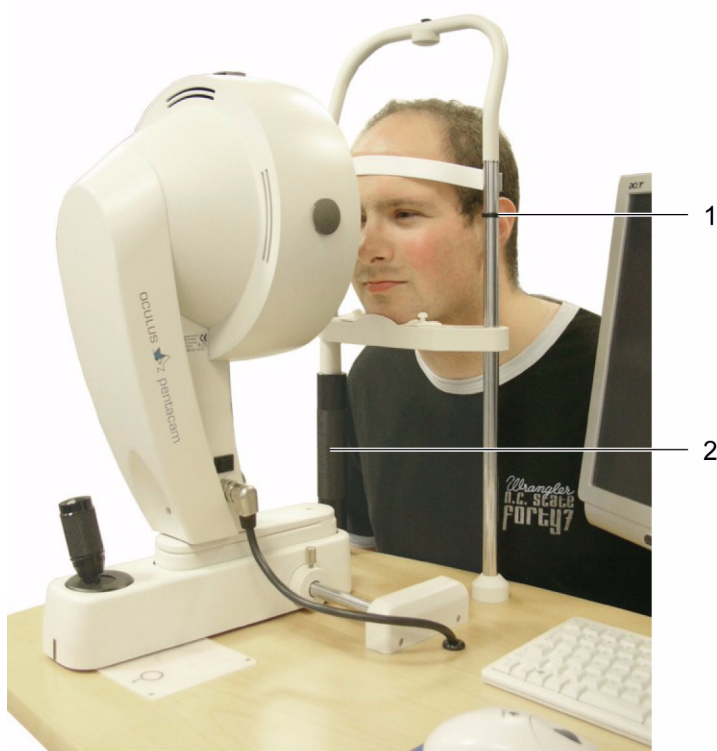
10 Measuring Procedure

Default settings

- ➔ Start the Scan menu ([sect. , page 10](#)).
- ➔ If necessary make changes to the image options for the particular part of the front of the eye that is to be examined.
- ➔ The default settings in the "3D Scan" options are "25 images/1 second".
- ➔ Adjust the table height.
- ➔ Ask the patient to place his or her head on the chin and forehead rest.

Rough adjustment

- ➔ The black ring marking between the chin rest and the forehead rest (1) should be used for gauging the required height of the patient's eyes.
 - ➔ Adjust the patient's eye level using the twist grip (2).
- The patient is positioned correctly when chin and forehead touch the rests and the eyes are level with the marking.

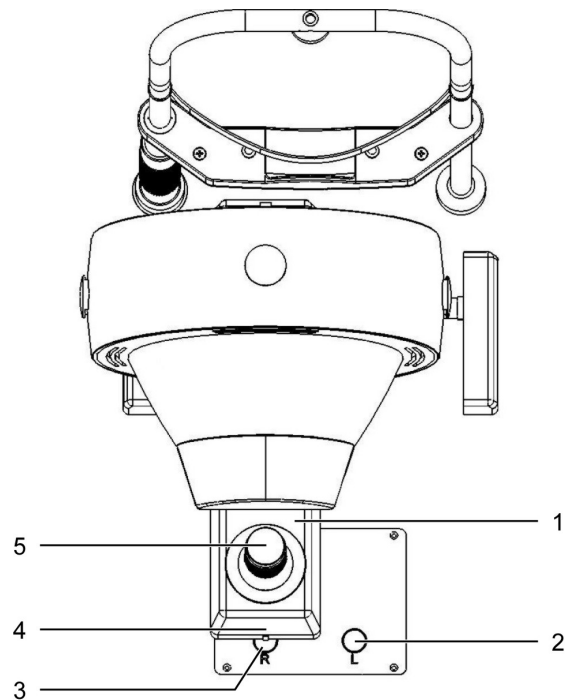


1 Marking (black ring)

2 Twist grip

Fig. 10-1: Rough adjustment of the chin and forehead rest

- ➔ Example of a rough adjustment for the right eye: Move the cross slide (1) until the marking at the end of the cross slide (4) roughly coincides with the circle R (3) on the sliding plate.



- 1 Cross slide
- 2 Left marking
- 3 Right marking

- 4 Marking on the cross slide
- 5 Joystick

Fig. 10-2: Markings on the cross slide

- ➔ Look at the patient's eye you are examining from one side and make sure that the blue slit light illuminates the cornea.
- ➔ If necessary, adjust the position of the cross slide to the left or right.

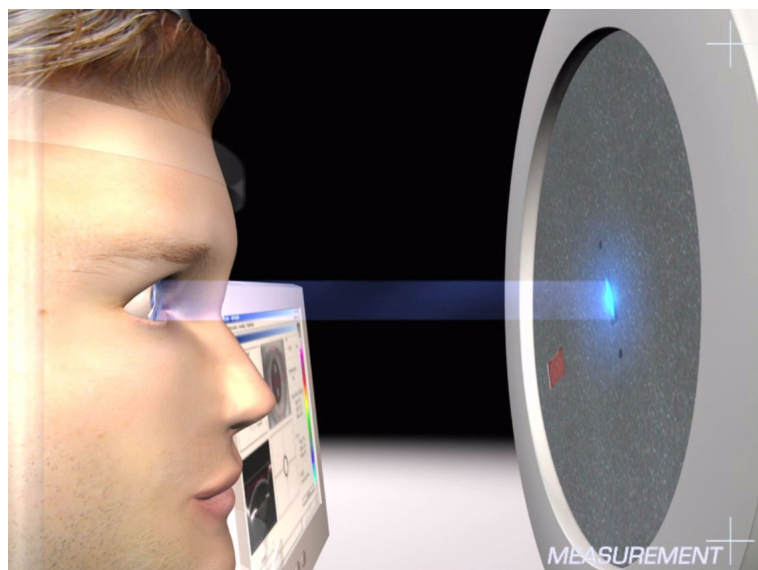


Fig. 10-3: Slit light on the cornea

Note

If the blue slit light is not visible, ensure that you have activated the [Slit Light] checkbox on the "Scan" screen.



Darkening the room / dark sheet

- If the lighting in the examination room has not been turned down or switched off, use the dark sheet supplied to cover the patient and the Pentacam.

Adjustment

- Move the cross slide towards the patient until the Scheimpflug image shows the cornea of the eye that you are examining.

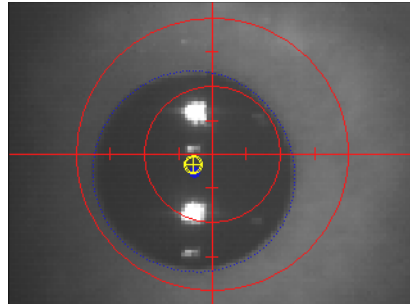
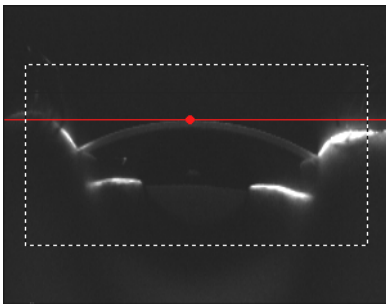


Fig. 10-4: Scheimpflug image (left) and pupil image (right)

The image is sharpest when the red dot coincides with the red line in the Scheimpflug image.

- Focus the pupil image by moving the joystick towards the Pentacam or away from it.
- Adjust the left/right position of the Pentacam and its height setting.
Move the joystick to the left or right and rotate the joystick clockwise or anticlockwise.

The tentative final position of the camera is reached when the yellow dot is in the centre of the crosshairs.

- Ask the patient to widen his or her eye.

Fine adjustment

- Make any fine adjustments required based on the information in the adjustment window. To do so, move the joystick in the specified directions.
Example: Move the joystick to the right and forwards (towards the patient) and also anticlockwise.

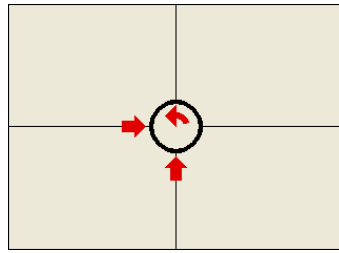


Fig. 10-5: Adjustment window

When you have achieved a sufficiently accurate position, a black cross will appear in the centre of the ring, surrounded by four black lines. The Pentacam will automatically begin measuring, alternately you can start the measuring procedure manually.

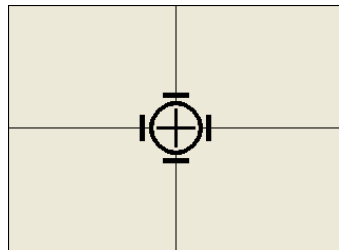


Fig. 10-6: Adjustment window once the final position has been reached

- For measuring manually: Start measuring by clicking [Start Scan] or by pressing the return key.



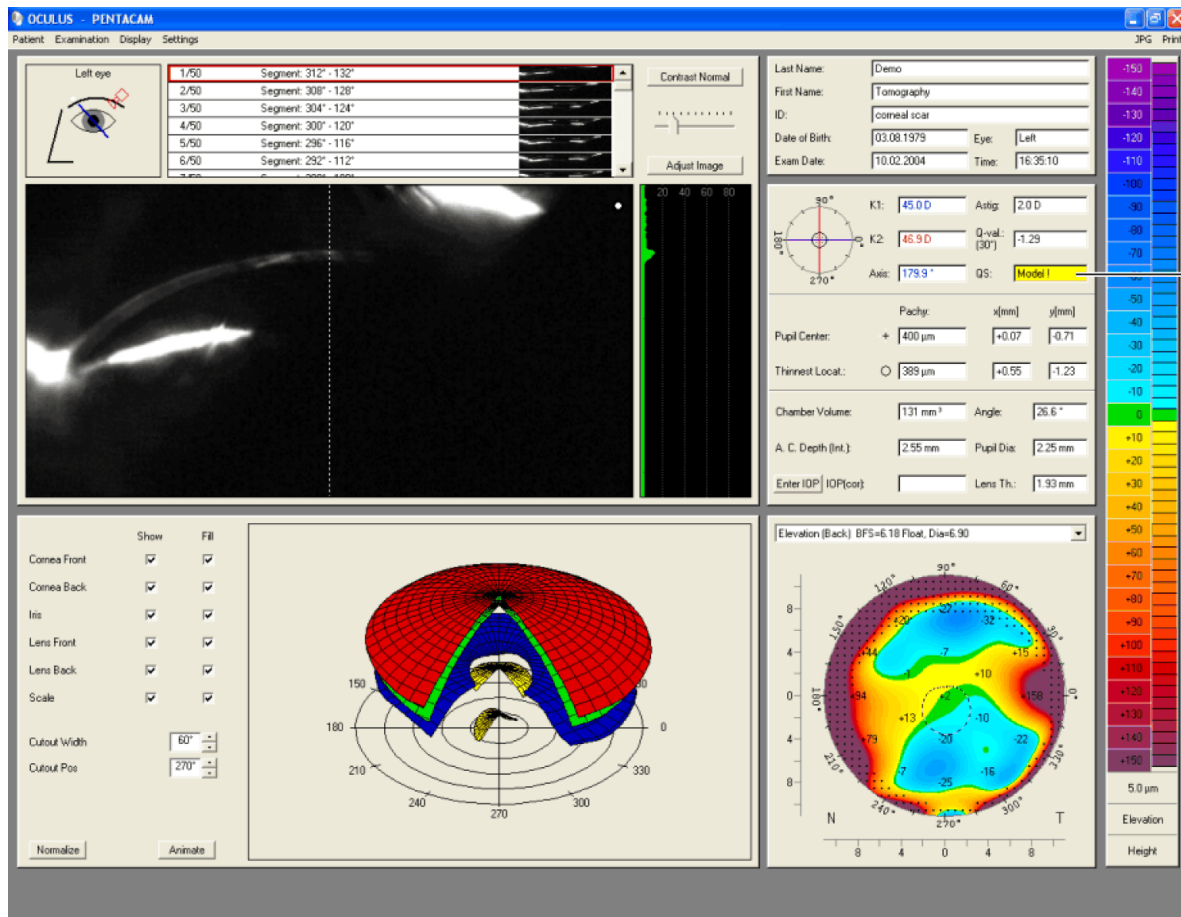
Attention

It may not be possible to carry out a measurement with manual release.

- Ask the patient to remove his or her head from the rest.
- Check the measurement results by referring to the quality specifications ([sect. 11, page 19](#)).

1.1 Quality Specifications

After you have begun measuring either automatically or manually, the Pentacam program opens. The "QS" button appears in a field below the patient data.

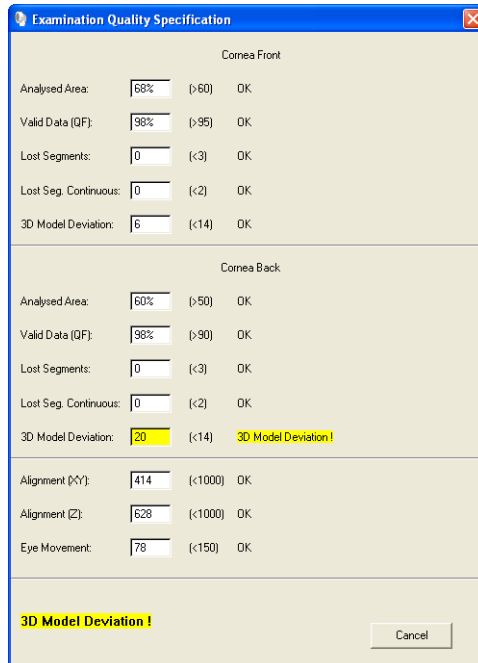


1 "QS" button

Fig. 11-1: Pentacam program with "QS" button

- ➔ If the "QS" button reads OK, the measurement is correct and can be reproduced.
- ➔ If the "QS" button is red, you must repeat the measurement.

- Click on the "QS" button if it is highlighted in yellow.
The following dialog box appears:



The dialog box titled "Examination Quality Specification" displays measurement results for Cornea Front and Cornea Back. The "3D Model Deviation" for Cornea Back is highlighted in yellow with a warning icon and the text "3D Model Deviation!".

Cornea Front			
Analysed Area:	68%	(>60)	OK
Valid Data (QF):	98%	(>95)	OK
Lost Segments:	0	(<3)	OK
Lost Seg. Continuous:	0	(<2)	OK
3D Model Deviation:	6	(<14)	OK

Cornea Back			
Analysed Area:	60%	(>50)	OK
Valid Data (QF):	98%	(>90)	OK
Lost Segments:	0	(<3)	OK
Lost Seg. Continuous:	0	(<2)	OK
3D Model Deviation:	20	(<14)	3D Model Deviation!
Alignment (XY):	414	(<1000)	OK
Alignment (Z):	628	(<1000)	OK
Eye Movement:	78	(<150)	OK

3D Model Deviation !

Cancel

Fig. 11-2: Examination Quality Specification

- Check the measurement results.
→ If you have any doubts, repeat the measurement.



Note

All examinations are automatically saved, regardless of the quality of the measurement taken.

Notes on individual parameters

- **Analysed Area**
If this value is less than the permissible threshold, the patient must widen his or her eye.
- **Valid Data**
If this value is less than the permissible threshold, you must darken the room.
- **Lost Segments and Lost Seg. Continuous**
If one of these values exceeds the permissible threshold, you must ask the patient not to blink while you are measuring.
- **Alignment (XY) and Alignment (Z)**
If one of these values exceeds the permissible threshold, you may have moved the cross slide at the moment you began measuring.
- **Eye Movement**
If this value exceeds the permissible threshold, it is possible that fixation of the patient is inadequate.

Terminating "QS"

- ➔ Click [Cancel] to return to the Pentacam program.
- ➔ If required, delete the measurement if the image is inadequate.
- ➔ Terminate the current examination which has been saved.
- ➔ If required, make preparations to examine another patient. In the "Pentacam" over-view, select the "Examination" menu and click [New Patient/End].

12 Transferring Measurement Results

Note

- Saving patient and examination data in a different directory.
- Transferring measurement data to another storage medium (e.g. USB drive).
- Sending stored data by e-mail.



12.1 Exporting patient data

- ➔ In Patient Data Management (*fig. 7-1, page 8*), select the patient you require. You can select a patient's entire data master or individual examinations.
 - ➔ Click [Export].
- The following dialog box appears:

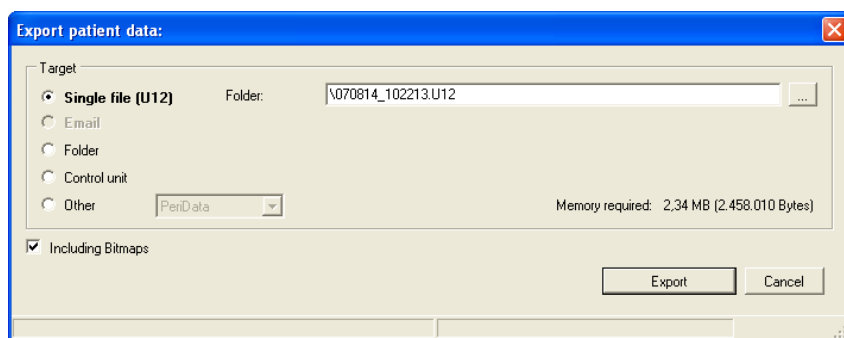


Fig. 12-1: Exporting patient data

- ➔ Select "Single File (U12)".
 - ➔ You can open the "Directory" by clicking the  button.
 - ➔ Specify the name and destination of the file you are saving.
 - ➔ Make sure you have selected [Including Bitmaps].
 - ➔ Click [Export].
- The patient and examination data have now been saved at the destination specified. You can send data stored on the hard drive as an e-mail attachment.



Note

Requirements for transferring data to another PC:

- A software licence purchased from an authorised dealer. This licence includes installation instructions and a Licence-Key.
- The Pentacam program must be installed on the other PC. If the program is updated on the Pentacam PC (sender), the program on the other PC (recipient) must also be updated.

12.2 Importing patient data

Import the patient and examination data you have received into your Pentacam software.

➔ In Patient Data Management (*fig. 7-1, page 8*), click [Import]

The following dialog box appears:

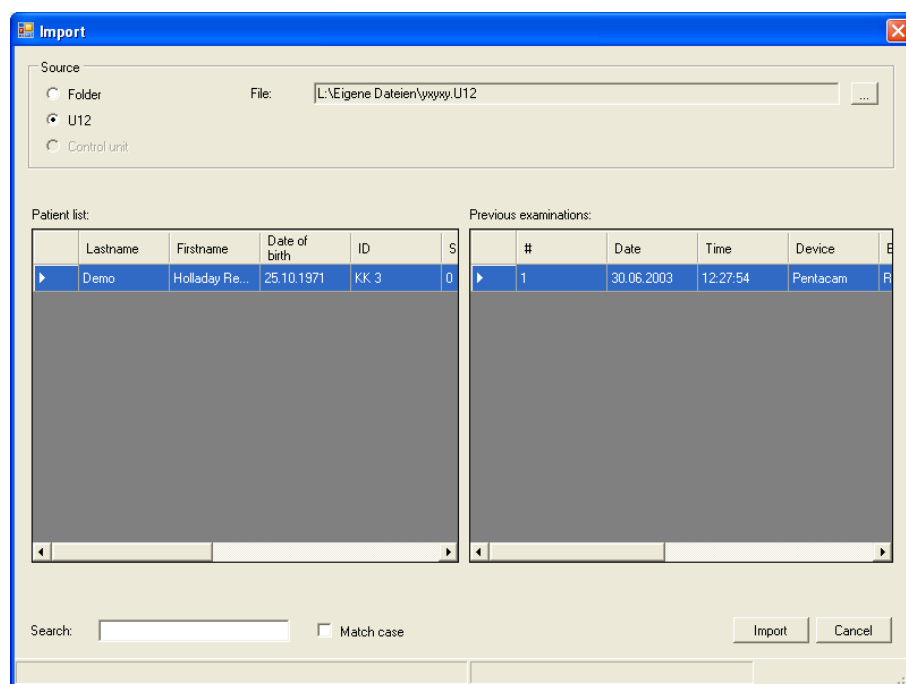


Fig. 12-2: Import

- ➔ Select the option "U12".
- ➔ Click the  button under "File".
- ➔ Look for the exported file in the selected folder.
- ➔ Open the exported file.
- ➔ Click [Import].

The imported files are marked in the Patient Data Management menu.

13 Test measurements with the Pentacam

The Pentacam/Pentacam HR is tested and calibrated in the Oculus factory. However, OCULUS Optikgeraete GmbH recommends to perform regular test measurements of the Pentacam/Pentacam HR.

A human eye should be measured initially. Minimum 5 consecutive measurements per eye should be done and the arithmetic mean has to be calculated and recorded.

Once in a month this measurement should be repeated using this eye as described above.

The arithmetic mean of the initial and the current measurement should be compared.

The tolerance range of the original and the current results are described in the table chart below.

	Tolerance Range
Curvature	+/- 0.25 dpt
Pachymetry	+/- 10 µm

If the difference between the initial and the current measurement is outside the tolerance range, please inform us or contact your authorized Oculus distributor to get further support.

14 Troubleshooting

Attention

If an error occurs which you are unable to correct by following the instructions below, label the device as "out of order" and contact our service department or an authorised dealer.



Error: After you have started the Pentacam program ([sect. 7.4, page 10](#)), the following dialog box appears: "No communication with Pentacam!".

Solution:

- Check whether the indicator light on the power supply is on. If not, connect the power supply to the mains.
- Check whether the power supply cable is correctly attached to the Pentacam.
- Check whether the blue slit light is visible in the Scan menu ([sect. , page 10](#)).
- Check whether the USB connector is properly inserted.
- Switch the Pentacam off and restart the PC. Switch the Pentacam on as soon as Patient Data Management becomes active. When you start the Pentacam program, the message, "Load Bootloader" must appear.
- Contact the service department or your authorised dealer.

15 Cleaning and Care

15.1 Instructions for cleaning



Attention

Always disconnect the power supply before cleaning.

- Never use cleaning agents that are aggressive, contain chlorine or solvents, or are rough or abrasive.

15.2 Sterilising and disinfecting



Note

No sterilising or disinfecting is required.

15.3 Cleaning



Attention

When cleaning the illuminated slit, do not use a cloth or other cleaning agents.

- Carefully clean the centre of the illuminated slit with purified compressed air.



Attention

The optical components illuminating the slit, and the lens in front of the camera are precision parts and sensitive to pressure. The surfaces of these components are susceptible to scratching.

- Clean the lens in front of the camera using a dry, lint-free cloth.
- It is best to clean the housing surfaces with a soft cloth and an antistatic cleaning agent.
- Wipe any residual particles using a mixture containing equal parts of spirits and distilled water, and add a dash of household washing up liquid.

16 Terms of Warranty and Servicing

16.1 Terms of warranty

The Pentacam you have purchased is a high-quality OCULUS product. This Pentacam was carefully manufactured using quality materials and modern production methods. Any software included in the delivery was tested by us and complies with technical standards. Prior to and while operating the device it is important that you observe the instruction manual and safety instructions.

The Pentacam carries a warranty to which you are entitled in accordance with the legal provisions.

If any unauthorised persons interfere with the Pentacam, all warranty entitlements shall be void. Any inappropriate modifications or repairs can cause grave danger to the user and patient.

Any entitlement to a warranty shall also be void if unauthorised persons interfere with the PC hardware and software supplied.

In the event of transport damage, we request that you notify the shipping company immediately and have the damage confirmed on the consignment note, to enable a proper claims settlement procedure.

Overall, the general terms and conditions of business and delivery apply as per the date of purchase.

16.2 Assumption of liability for functions and damage

OCULUS shall only consider itself responsible for the safety, reliability and usability of the Pentacam if the Pentacam is used in accordance with this instruction manual.

There are no user-serviceable parts either on or inside the Pentacam.

OCULUS shall not assume any liability if assembly, extensions, adjustments, changes or repairs are carried out by unauthorised personnel, if the Pentacam is maintained improperly or if it is handled incorrectly.

If the work described above is carried out by persons authorised to do so, they must be requested to supply documentation detailing the nature and scope of repairs, and if applicable to specify modifications to the rated data or area of work. This certificate must bear a date, a signature, specify who carried out the work, and contain company information.

On request, and for this purpose, OCULUS will supply authorised persons with spare parts lists and additional descriptions.

Only OCULUS original spare parts may be used for repairs.

16.3 Address of the manufacturer and service department

Our service department or authorised representatives will furnish you with additional information.

Address of the manufacturer and service department:

OCULUS Optikgeräte GmbH
Münchholzhäuser Straße 29
35582 Wetzlar, Germany
Tel.: 0641/2005-0
Fax: 0641/2005-255
E-mail: sales@oculus.de
www.oculus.de



USA:

OCULUS Inc.
2125 196th Street SW
Suite 112
Lynnwood, WA 98036
Tel.: 425-670-9977
E-mail: oculusincusa@msn.com



17 Declaration of Conformity



KONFORMITÄTSERKLÄRUNG /
DECLARATION DE CONFORMITE /
DECLARATION OF CONFORMITY /
DICLARATIONE DE CONFORMITA /
DECLARATION DE CONFORMITA

OCULUS Optikgeräte GmbH
Münchholzhäuser Str.29
D-35582 Wetzlar

Tel: ++49 641 / 20 05 - 0
Fax: ++49 641 / 20 05 - 255

Wir / Nous / We / Noi / Nosotros

OCULUS Optikgeräte GmbH

Erklären in alleiniger Verantwortung, dass das Medizinprodukt

déclarons sous notre propre responsabilité que le dispositif médical
declare on our own responsibility that the medical device
dichiaro sotto propria responsabilità che il dispositivo medico
declaramos en sola responsabilidad que el producto medicina

Name / nom / name / nome / nombre

Pentacam / Pentacam HR

Typ / type ou modèle / type or model /
tipo o modello / Tipo y modelo

70700 / 70900

allen Anforderungen der Richtlinie 93/42/EWG entspricht.

remplit toutes les exigences de la 93/42/CEE qui le concernait.
meets all the provisions of the Directive 93/42/EEC which apply to him.
adempie a tutte le esigenze della Direttiva 93/42/CEE che lo riguardano.
reuna todos los requisitos de la norma 93/42/CEE

Einstufung: (Richtlinie 93/42/EWG, Anhang IX)

Classification: (MDD 93/42/EEC, annex IX)
Classification: (MDD 93/42/EEC, annex IX)
Classification: (MDD 93/42/EEC, annex IX)
Clasificación: (MDD 93/42/EEC, annex IX)

Ila

Konformitätsbewertung nach: Richtlinie 93/42/EWG, Anhang VII

Conformity according: MDD 93/42/EEC, annex VII
Conformité en: MDD 93/42/EEC, annex VII
Conformità a: MDD 93/42/EEC, annex VII
Conformidad para : MDD 93/42/EEC, annex VII

Angewandte harmonisierte Normen:

Normes harmonisées appliquées:
Applied harmonized standards:
Norme armonizzate applicate
Norma armonizaciónada

DIN EN 60601-1 DIN EN 14971
DIN EN 60601-1-2
DIN EN ISO 15004

Ort, Datum /

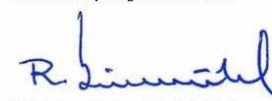
lieu, date / place, date / luogo, data / lugar, fecha

Name und Funktion /

nom et fonction / name and function /
nome e funzione / nombre y function

Geschäftsführer / Managing Director
OCULUS Optikgeräte GmbH

Wetzlar, 29.08.2006


Dipl.-Ing. Rainer Kirchhübel

18 Technical Specifications / Minimum PC Requirements

Measuring equipment

Camera	Digital CCD camera
Light source	Blue LEDs (475 nm, UV-free)
Speed	50 images in 2 seconds, 500 measuring points (HR: 2760) per recorded image
Number of evaluated measuring points	Max. 25,000 (HR: max. 138,000)
Dimensions H x W x D	535 x 280 x 360 mm
Weight	9 kg

Measuring range

Curvature:	3 to 38 mm 9 to 99 dpt
Accuracy	± 0.2 dpt
Reproducibility	± 0.2 dpt
Working distance	80 mm

Power supply

Mains connection	100 - 240 V AC, 50 - 60 Hz
Power input	60 W
Power output	24 V, 2.1 A
Fuses	Integrated overcurrent shut-off

Other information

CE marking	CE 0123
Contraindications	None noted
Lifecycle expectancy	Up to 10 years

Classification according to IEC 60601 - 1

Type of protection against electrical shock	Protection class 1
Level of protection against electrical shock	Type II B
Level of protection against damaging water entry	IP20

Ambient operating requirements

Temperature	+10 °C to +40 °C
Humidity	30% to 75%
Air pressure	700hPa to 1060hPa

Transport and storage requirements (according to IEC 601 - 1)

Ambient temperature	-40 °C to +70 °C
Relative humidity, including condensation	10% to 100%
Air pressure	500hPa to 1060hPa

Minimum PC requirements

CPU	Pentium Intel Centrino 1.4 GHz Pentium Intel Pentium 2.4 GHz
Operating system	Windows™ XP SP2 Microsoft.NET Framework, version 2.0
RAM	1 GB RAM
Graphics card	ATI Radeon 9200 SE 128 MB